

# AI Assistant Coding

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Batch-23

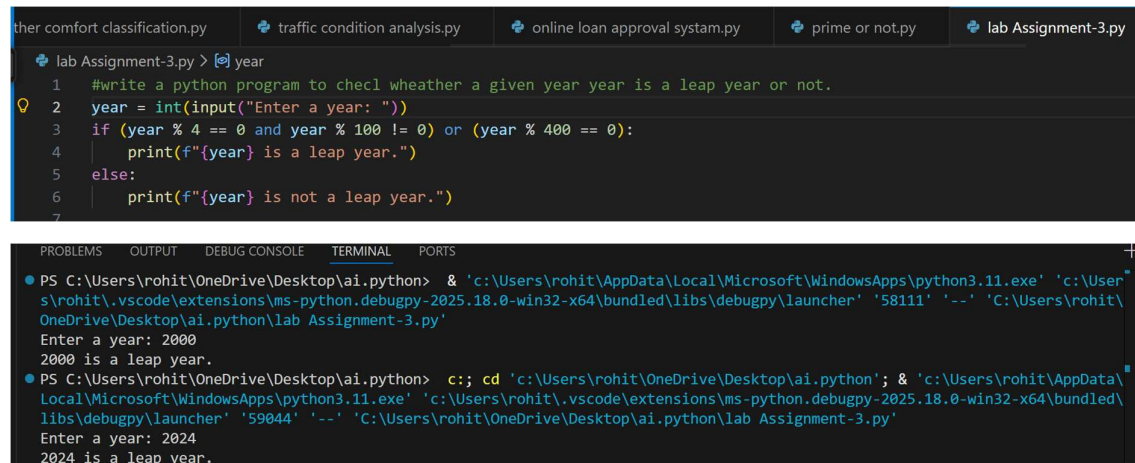
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## Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.



The screenshot shows a VS Code editor with a file named 'lab Assignment-3.py'. The code is a Python function that checks if a year is a leap year. The code is as follows:

```
1 #write a python program to check wheather a given year year is a leap year or not.
2 year = int(input("Enter a year: "))
3 if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
4     print(f"{year} is a leap year.")
5 else:
6     print(f"{year} is not a leap year.")
7
```

The terminal output shows the execution of the script. The user enters '2000' and the output is '2000 is a leap year.'. The user then enters '2024' and the output is '2024 is a leap year.'.

## Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency.

```
#generate a python program to find the gcd of two numbers :12 and 18
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a
num1 = 12
num2 = 18
print(f"The GCD of {num1} and {num2} is {gcd(num1, num2)}")
```

```
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c:: cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63109' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
GCD of 12 and 18 is 6
```

### Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases.

```
21 #generate a python code to find LCM of two numbers
22 """
23 examples:
24 input:4 and 6
25 output:12
26 input:5 and 10
27 output:10
28 input:7 and 3
29 output:21
30 """
31 def lcm(a, b):
32     def gcd(x, y):
33         while y:
34             x, y = y, x % y
35         return x
36     return abs(a * b) // gcd(a, b)
37 num1 = int(input("Enter first number: "))
38 num2 = int(input("Enter second number: "))
39 print(f"The LCM of {num1} and {num2} is {lcm(num1, num2)}")
40
```

```

PS C:\Users\rohit\OneDrive\Desktop\ai.python> c:: cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '53505' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter first number: 4
Enter second number: 6
LCM of 4 and 6 is 12
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c:: cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '50934' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter first number: 5
Enter second number: 10
LCM of 5 and 10 is 10
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c:: cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '50951' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter first number: 7
Enter second number: 3
LCM of 7 and 3 is 21

```

#### Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

- Test with valid and invalid binary inputs.
- Identify missing validation logic.

```

40 #generate a python code that converts a binary number into decimal
41 binary_num = input("Enter a binary number: ")
42 decimal_num = int(binary_num, 2)
43 print(f"The decimal equivalent of binary {binary_num} is {decimal_num}")
44

```

```

PS C:\Users\rohit\OneDrive\Desktop\ai.python> c:: cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '50898' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a binary number: 101
The decimal equivalent of binary 101 is 5.

```

#### Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

```

45 #generate a python code that converts a decimal number into binary number
46 """
47 Example:
48 input:10
49 output:1010 """
50 decimal_num = int(input("Enter a decimal number: "))
51 binary_num = bin(decimal_num).replace("0b", "")
52 print(f"The binary equivalent of decimal {decimal_num} is {binary_num}")

```

```

PS C:\Users\rohit\OneDrive\Desktop\ai.python> c;; cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56327' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a decimal number: 1010
The binary equivalent of decimal 1010 is 1111110010.
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c;; cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62776' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a decimal number: 10
The binary equivalent of decimal 10 is 1010.

```

### Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

```

54 #generate a python code to find whether the given numver is harsad number or not
55 """
56 examples
57 Input: 18
58 Output: Harshad Number
59
60 Input: 21
61 Output: Harshad Number
62
63 Input: 19
64 Output: Not a Harshad Number"""
65 num = int(input("Enter a number: "))
66 sum_of_digits = sum(int(digit) for digit in str(num))
67 if num % sum_of_digits == 0:
68     print(f"{num} is a Harshad Number")
69 else:
70     print(f"{num} is Not a Harshad Number")

```

```
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c::; cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50031' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a number: 18
18 is a Harshad number.
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c::; cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50052' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a number: 45
45 is a Harshad number.
PS C:\Users\rohit\OneDrive\Desktop\ai.python> c::; cd 'c:\Users\rohit\OneDrive\Desktop\ai.python'; & 'c:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\rohit\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50075' '--' 'C:\Users\rohit\OneDrive\Desktop\ai.python\lab Assignment-3.py'
Enter a number: 98
98 is not a Harshad number.
```